

Multivariate Normal Distribution, Copulas, and Density Estimation

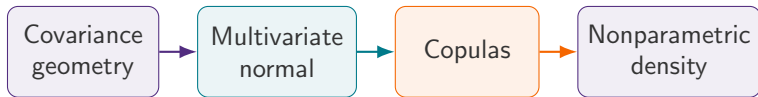
From covariance geometry to flexible multivariate modeling

Whitney Huang

Clemson University | DSA 8070

Readings: Zelterman (2015), Chs. 5–7; Izenman (2008),
Secs. 4.1, 4.3, 4.5

Roadmap: From Matrix Algebra to Distribution Modeling



1. Use the multivariate normal to connect probability with covariance geometry
2. Separate marginal behavior from dependence using copulas
3. Relax parametric assumptions using kernel density estimation

The common question: how should we represent a multivariate distribution?

Learning Objectives

By the end of this lecture, you should be able to:

- ▶ interpret the mean vector and covariance matrix of a multivariate normal distribution;
- ▶ use linear transformations and conditional distributions of multivariate normal vectors;
- ▶ connect Mahalanobis distance, eigenvalues, and probability ellipsoids;
- ▶ explain how a copula separates marginal distributions from dependence;
- ▶ explain how kernel density estimation provides a nonparametric alternative to a specified density family.

Why the Multivariate Normal?

Statistical role

The multivariate normal is a basic reference model for a vector of jointly varying measurements.

- ▶ Mathematically tractable
- ▶ Determined by μ and Σ
- ▶ Closed under linear transformations and marginalization

Asymptotic role

Multivariate central limit theorems make normal approximations natural for many vector-valued estimators.

- ▶ Sample mean vectors
- ▶ Regression estimators
- ▶ Many likelihood-based estimators

Key idea

The covariance matrix controls both *dependence* and *geometry*.

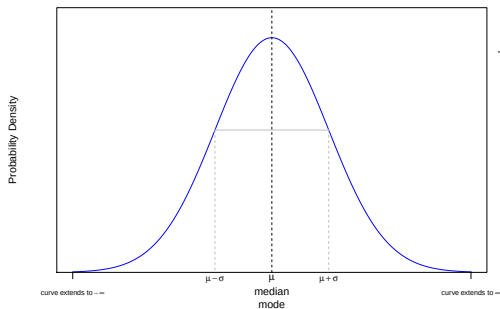
Start with the Univariate Normal

For $X \sim N(\mu, \sigma^2)$,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2} \right\}.$$

The exponent contains

$$\frac{(x - \mu)^2}{\sigma^2} = (x - \mu)(\sigma^2)^{-1}(x - \mu)$$



- ▶ center by μ ;
- ▶ scale by variance;
- ▶ measure squared statistical distance.

The Multivariate Normal Density

Let $\mathbf{X} \in \mathbb{R}^p$ with mean vector $\boldsymbol{\mu}$ and positive-definite covariance matrix Σ . We write

$$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma).$$

Its density is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}.$$

Location

$\boldsymbol{\mu}$ identifies the center of the distribution.

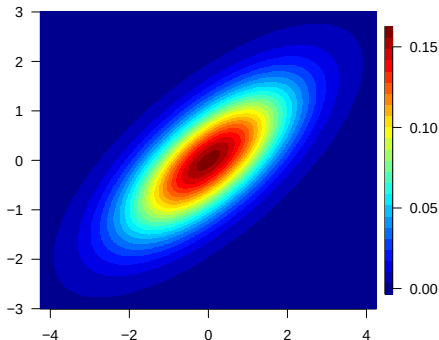
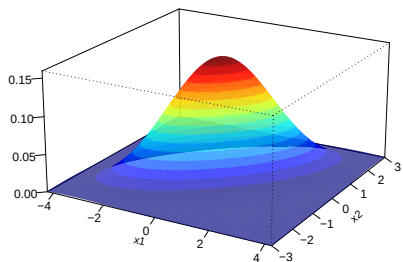
Shape and scale

Σ determines marginal variances, dependence, orientation, and spread.

Compare the exponent with the Mahalanobis distance from the matrix-algebra lecture.

An Example of a Bivariate Normal Density

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \right)$$



Multivariate Central Limit Theorem

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be i.i.d. p -dimensional random vectors with

$$E(\mathbf{X}_i) = \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{X}_i) = \Sigma.$$

Under standard regularity conditions,

$$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \Sigma),$$

or equivalently, for large n ,

$$\bar{\mathbf{X}} \dot{\sim} N_p\left(\boldsymbol{\mu}, \frac{\Sigma}{n}\right).$$

Key idea

The population itself need not be multivariate normal for the **sample mean vector** to be approximately multivariate normal.

Core Properties of the Multivariate Normal

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$, then:

1. **Marginals are normal.** Any subvector of $\mathbf{X}$ is multivariate normal.
2. **Linear transformations are normal.** For $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$,

$$\mathbf{Y} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\Sigma\mathbf{A}^T).$$

In particular, $Y = \mathbf{a}^T \mathbf{X}$ is univariate normal with variance $\mathbf{a}^T \Sigma \mathbf{a}$.

3. **Conditionals are normal.** After partitioning $\mathbf{X}$, conditioning one block on another preserves normality.

Where this appears later

Regression, missing-data prediction, Gaussian processes, and discriminant analysis.

Example: A Linear Combination of Cholesterol Measurements

Suppose X_1, X_2, X_3 are cholesterol measurements at days 0, 2, and 4, with

$$\bar{\mathbf{x}} = \begin{bmatrix} 259.5 \\ 230.8 \\ 221.5 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 2276 & 1508 & 813 \\ 1508 & 2206 & 1349 \\ 813 & 1349 & 1865 \end{bmatrix}.$$

We are interested in the change from day 0 to day 2:

$$\Delta = X_2 - X_1 = \mathbf{a}^T \mathbf{X}, \quad \mathbf{a} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Use the matrix-algebra result

$$E(\Delta) = \mathbf{a}^T \boldsymbol{\mu}, \quad \text{Var}(\Delta) = \mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a}.$$

Cholesterol Example: Mean and Variance of Δ

Using the sample summaries,

$$\widehat{E}(\Delta) = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 259.5 \\ 230.8 \\ 221.5 \end{bmatrix} = -28.7,$$

and

$$\begin{aligned} \widehat{\text{Var}}(\Delta) &= \mathbf{a}^T S \mathbf{a} = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2276 & 1508 & 813 \\ 1508 & 2206 & 1349 \\ 813 & 1349 & 1865 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -768 & 698 & 536 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \\ &= 1466 \end{aligned}$$

Therefore, under a multivariate normal model,

$$\Delta \sim N(-28.7, 1466)$$

Partitioning Leads Naturally to Conditional Distributions

Partition

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}.$$

If $\mathbf{X}$ is jointly multivariate normal, then

$$\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim N \left(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} \right).$$

Conditional mean

Observed $\mathbf{X}_2$ shifts our prediction of $\mathbf{X}_1$.

Conditional covariance

Conditioning reduces uncertainty when the blocks are associated.

From the Density to Elliptical Contours

The multivariate normal density depends on $\mathbf{x}$ through

$$d_M^2(\mathbf{x}, \boldsymbol{\mu}) = (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}).$$

Therefore, points satisfying

$$(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$$

have the same density.

- ▶ In two dimensions: ellipses.
- ▶ In three dimensions: ellipsoids.
- ▶ In p dimensions: ellipsoidal level sets.

Key idea

The inverse covariance converts the raw coordinate system into a covariance-adjusted distance.

Probability Regions and χ^2 Distribution

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$, then

$$d^2 = (\mathbf{X} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2.$$

Thus a $100(1 - \alpha)\%$ probability region is

$$\{\mathbf{x} : (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \leq \chi_{p, 1-\alpha}^2\}.$$

For $p = 2$ and $1 - \alpha = .95$,

$$c = \sqrt{\chi_{2, .95}^2} \approx 2.448.$$

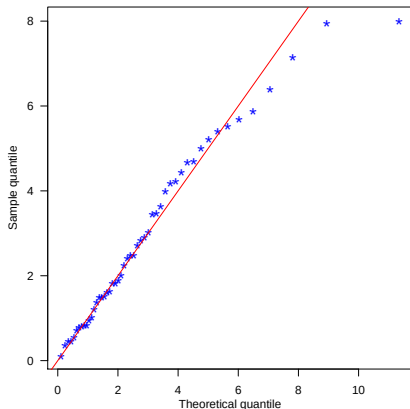
Takeaway: The squared Mahalanobis distance determines how far an observation is from the mean relative to the covariance structure. The 95% probability region contains all points whose Mahalanobis distance from $\boldsymbol{\mu}$ is at most 2.448

Multivariate Normality and Outliers

If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then

$$d^2 = (\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2.$$

This result can be used to assess **multivariate normality** and detect **outliers**



- ▶ Sort $(\mathbf{x}_i - \bar{\mathbf{x}})^T \mathbf{S}^{-1} (\mathbf{x}_i - \bar{\mathbf{x}})$ in an increasing order to get **sample quantiles**
- ▶ Calculate the **theoretical quantiles** using the **chi-square quantiles** with $p = \frac{i-0.5}{n}$, $i = 1, \dots, n$
- ▶ Plot sample quantile against theoretical quantiles

Eigenvalues Draw the Probability Ellipse

Suppose

$$\Sigma = P\Lambda P^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p).$$

For the contour

$$(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2,$$

the principal axes are determined by the eigenvectors $\mathbf{e}_j$, and the semi-axis lengths are

$$c\sqrt{\lambda_j}.$$

Hence the axis endpoints are

$$\boldsymbol{\mu} \pm c\sqrt{\lambda_j} \mathbf{e}_j.$$

Key idea

Large covariance eigenvalue $\Rightarrow$ more variation $\Rightarrow$ a longer probability-ellipse axis.

Example: Geometry from a Covariance Matrix

Let

$$\boldsymbol{\mu} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} 64 & 16 \\ 16 & 9 \end{bmatrix}.$$

The eigenpairs are approximately

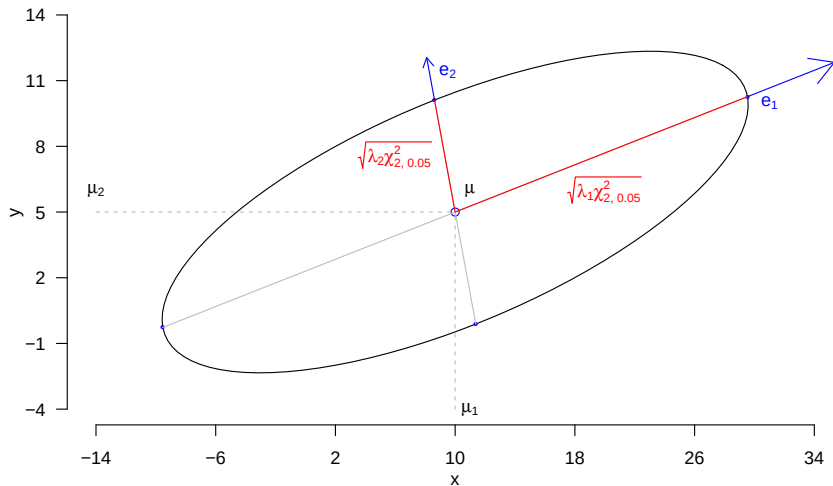
$$\lambda_1 = 68.316, \quad \mathbf{e}_1 = \begin{bmatrix} -0.9655 \\ -0.2604 \end{bmatrix}, \quad \lambda_2 = 4.684, \quad \mathbf{e}_2 = \begin{bmatrix} 0.2604 \\ -0.9655 \end{bmatrix}.$$

For the 95% probability ellipse, $c \approx 2.448$

- ▶ Major semi-axis length: $c\sqrt{\lambda_1} \approx 20.24$.
- ▶ Minor semi-axis length: $c\sqrt{\lambda_2} \approx 5.30$.

The eigenvectors give directions; the square roots of eigenvalues give standard-deviation scales

Graph of 95% Probability Contour



Applied Check: Wechsler Adult Intelligence Scale

For $n = 37$ subjects, suppose four test components are observed:

1. Information,
2. Similarities,
3. Arithmetic,
4. Picture Completion.

A useful workflow is:

1. compute $\bar{x}$ and S ;
2. inspect eigenvalues/eigenvectors to understand dominant directions;
3. examine Mahalanobis distances and a chi-squared Q-Q plot;
4. decide whether the multivariate normal is an adequate working model.

Beyond Normality: What Can Go Wrong?

The multivariate normal imposes several linked assumptions:

- ▶ every marginal distribution is normal;
- ▶ dependence is elliptical and summarized by correlation;
- ▶ Gaussian dependence has no asymptotic tail dependence when $|\rho| < 1$.

But in applications we may want, for example,

skewed margin + heavy-tailed margin + non-Gaussian dependence.

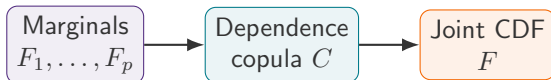
Copulas let us model the margins and dependence separately.

Copulas [Sklar, 1959; Joe, 1997]: Separate Margins from Dependence

Let F be a joint CDF with marginal CDFs $F_1, \dots, F_p$. Sklar's theorem gives

$$F(x_1, \dots, x_p) = C\{F_1(x_1), \dots, F_p(x_p)\},$$

where C is a copula: a multivariate CDF with Uniform(0, 1) margins.

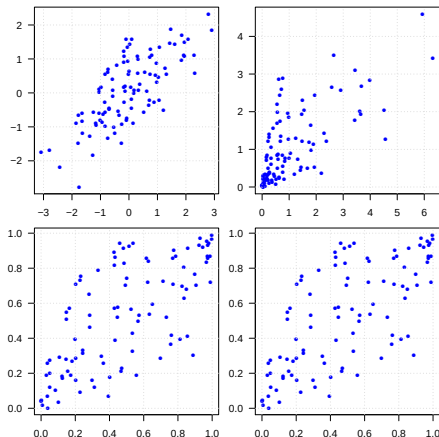


Gaussian Model versus Copula Construction

	Multivariate normal	Copula approach
Margins	Normal	Chosen separately
Dependence	Encoded by Σ / correlation	Encoded by copula C
Asymmetry	Limited	Possible with suitable copulas
Tail dependence	None for Gaussian copula when $ \rho < 1$	Possible with, e.g., t , Clayton, Gumbel copulas

A copula is not automatically “more flexible”; flexibility depends on the chosen copula family and marginal models.

Illustration: Same Copula, Different Margins

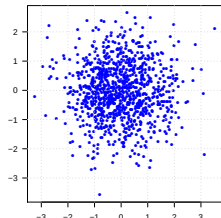


Interpretation

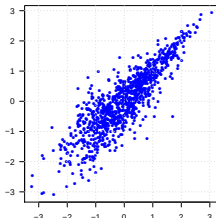
Keeping the copula fixed preserves the modeled dependence on the probability scale, while changing the marginal CDFs changes the shape and scale of each variable.

More Examples

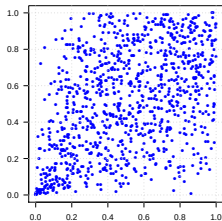
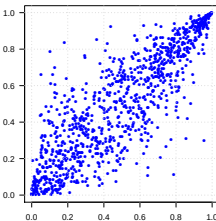
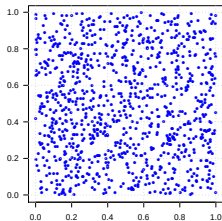
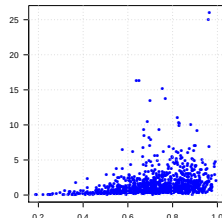
Marginal: normal
and normal **Copula:**
Gaussian $\rho = 0$



Marginal: normal
and normal **Copula:**
Gumbel $\theta = 3$



Marginal: Beta and
Log-normal **Copula:**
Clayton $\theta = 0.95$



⇒ The copula approach allows for more options for marginal and dependence modeling

A Practical Copula Modeling Workflow

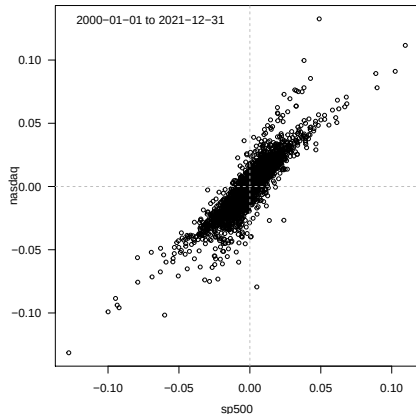
For observations (x_{1i}, x_{2i}) , $i = 1, \dots, n$:

1. **Model the margins:** estimate F_1 and F_2 .
2. **Transform to the unit square:** $u_{ji} = \hat{F}_j(x_{ji})$ (or use rank-based pseudo-observations).
3. **Model dependence:** fit a copula C to (u_{1i}, u_{2i}) .
4. **Recombine:** construct

$$\hat{F}(x_1, x_2) = \hat{C}\{\hat{F}_1(x_1), \hat{F}_2(x_2)\}.$$

5. **Validate both pieces:** check marginal fit and dependence fit separately.

Financial Illustration: S&P 500 and Nasdaq Returns



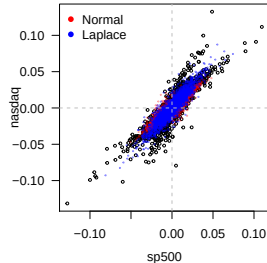
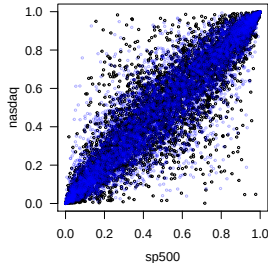
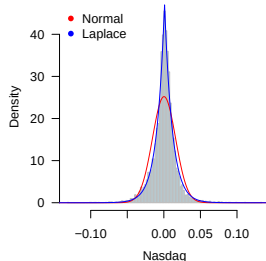
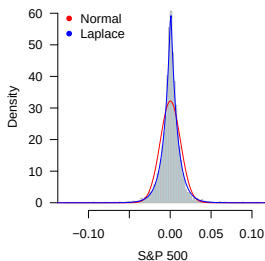
A possible fitted model combines:

- ▶ a marginal model for S&P 500 returns;
- ▶ a marginal model for Nasdaq returns;
- ▶ a copula for cross-market dependence.

Why this decomposition helps

Heavy-tailed marginal behavior and strong joint tail behavior need not be forced into one Gaussian covariance model.

Marginals, Copula, and Joint Distribution



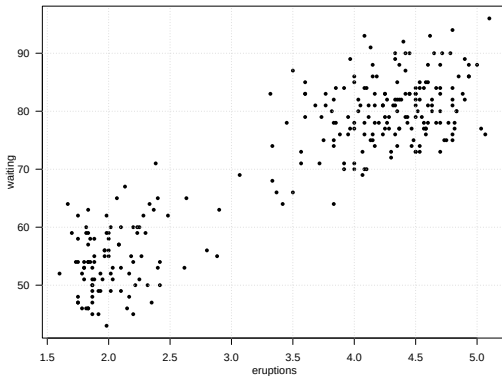
Marginals: Laplace

Copula: t -copula

When a Parametric Family Is Too Restrictive

The Old Faithful data contain two variables:

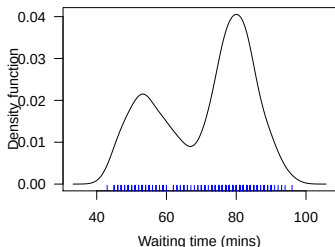
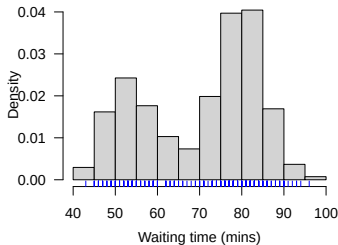
- ▶ eruption duration;
- ▶ waiting time until the next eruption.



A single multivariate normal density may miss multimodality or other non-elliptical structure.

Transition from Histogram to Kernel Density

Goal: to estimate the probability density function $f(x)$



► **Histogram:**

$$\hat{f}(x) = \sum_{j=1}^m \frac{\# \text{ of } x_i \in B_j}{nh} \mathbb{1}(x \in B_j),$$

where B_j is the j th bin and h is the binwidth

► **Kernel Density:**

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

where $K(\cdot)$ is the kernel function and h is the **bandwidth**

Multivariate Kernel Density Estimation

For p -dimensional observations $\mathbf{x}_1, \dots, \mathbf{x}_n$, a common form is

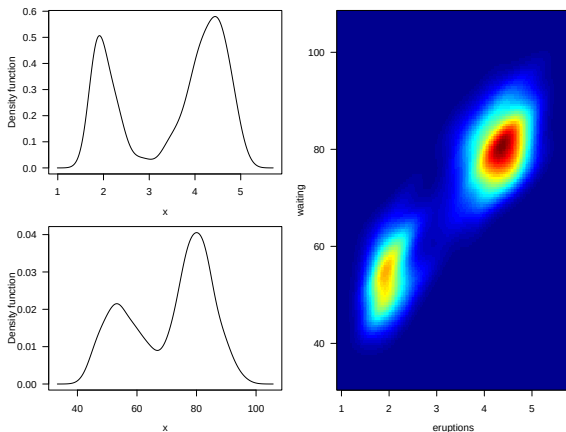
$$\hat{f}_H(\mathbf{x}) = \frac{1}{n|H|^{1/2}} \sum_{i=1}^n K\left(H^{-1/2}(\mathbf{x} - \mathbf{x}_i)\right),$$

where H is a positive-definite bandwidth matrix.

- ▶ $H = h^2 I$: same smoothing scale in every direction.
- ▶ Diagonal H : variable-specific smoothing.
- ▶ Full H : smoothing can adapt to orientation and dependence.

As dimension increases, density estimation becomes substantially harder: an example of the curse of dimensionality.

Bandwidth Controls the Bias–Variance Tradeoff



Bandwidth too small

Low smoothing; captures noise and produces a variable estimate.

Bandwidth too large

High smoothing; may erase genuine features such as multiple modes.

Three Ways to Model a Multivariate Distribution

	Multivariate normal	Copula model	KDE
Main structure	μ, Σ	Margins + copula	Local smoothing
Strength	Simple, interpretable geometry	Separates margins and dependence	Few parametric assumptions
Challenge	Restrictive shape/tails	Model choices for both pieces	Bandwidth; high dimension
Best when	Elliptical model is adequate	Dependence needs flexibility	Shape is difficult to specify

Model choice should follow the scientific structure and the inferential goal, not flexibility alone.

Concept Map: What Carries Forward?

Building block	Where it appears
Σ^{-1}	Mahalanobis distance, MVN density, classification
Eigenpairs of Σ	Probability ellipsoids, PCA, dimension reduction
Partitioned Σ	Conditional MVN, prediction, Gaussian processes
Marginal CDFs F_j	Probability integral transform, copulas
Copula C	Flexible dependence and joint-tail modeling
Bandwidth h or H	Nonparametric density estimation

Takeaways

- ▶ The multivariate normal connects matrix algebra with probability
- ▶ Mahalanobis distance defines density contours and probability regions
- ▶ Conditional normals update one set of variables given another
- ▶ Copulas separate marginal distributions from dependence
- ▶ Kernel density estimation provides a flexible nonparametric alternative